

ECO208 Macroeconomic Theory

Lecture 21

Sebastian Dyrda
University of Toronto Mississauga

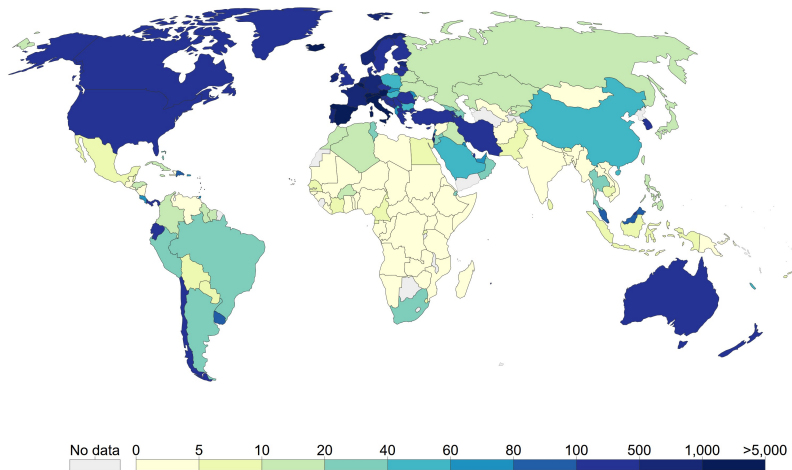
UTM, April 1, 2020

The COVID-19

Global Pandemic and Macroeconomic Policy

Total confirmed cases of COVID-19 per million people, Mar 31, 2020

The number of confirmed cases is lower than the number of total cases. The main reason for this is limited testing.

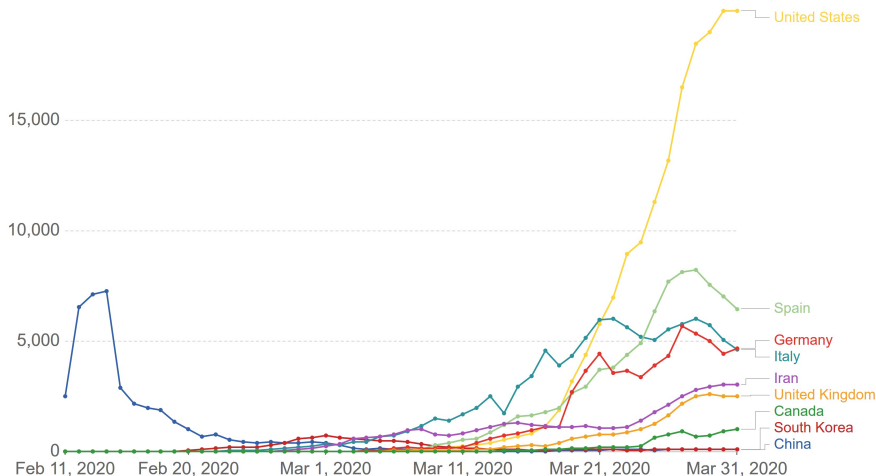


Source: European CDC – Situation Update Worldwide – Last updated 31st March, 13:00 (London time) [OurWorldInData.org/coronavirus](https://www.ourworldindata.org/coronavirus) • CC BY

Note: The large number of cases globally and in China on Feb 17 is the result of a change in reporting methodology.

COVID-19 – Daily new confirmed cases – rolling 3-day average

The number of confirmed cases is lower than the number of total cases. The main reason for this is limited testing.

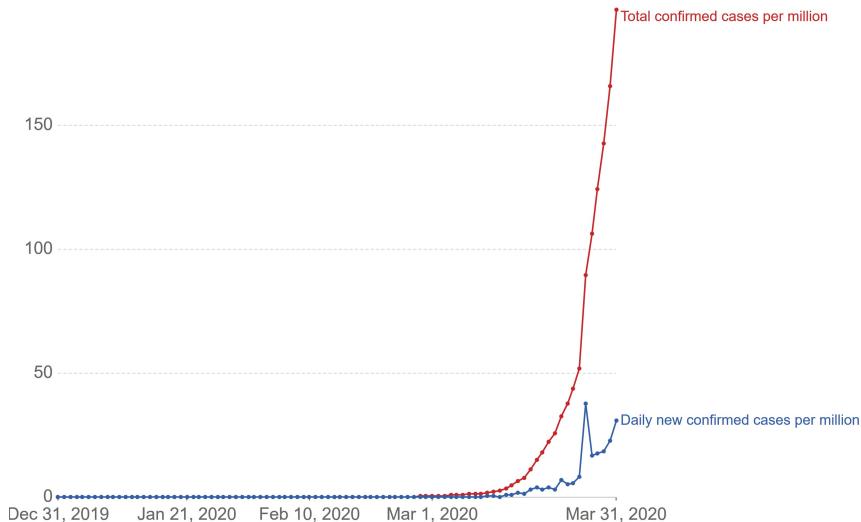


Source: European CDC – Situation Update Worldwide – Last updated 31st March, 13:00 (London time) OurWorldInData.org/coronavirus • CC BY

Note: The rolling average is the average across three days – the confirmed cases on the particular date, and those on the previous and the following day. For example, the value for 26th March is the average over the 25th, 26th and 27th March. The latest value is calculated as the average of confirmed cases on the particular date and the previous day.

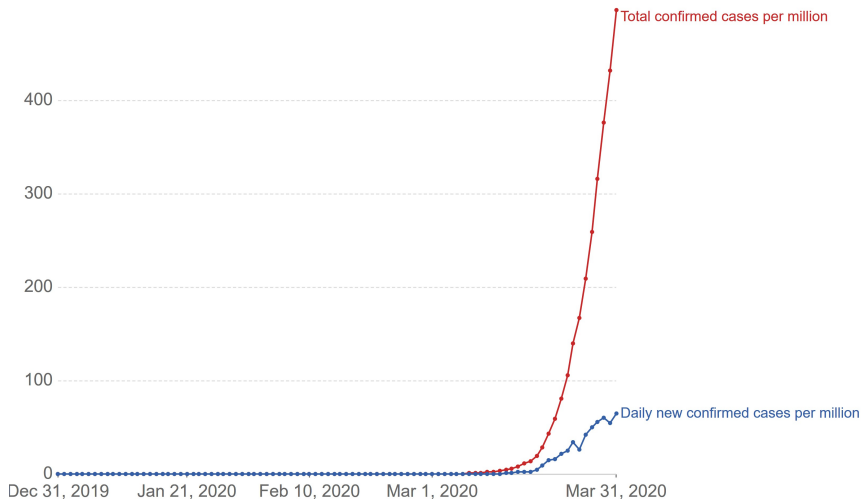
Total and daily confirmed COVID-19 cases per million people, Canada

The number of confirmed cases is lower than the number of total cases. The main reason for this is limited testing.



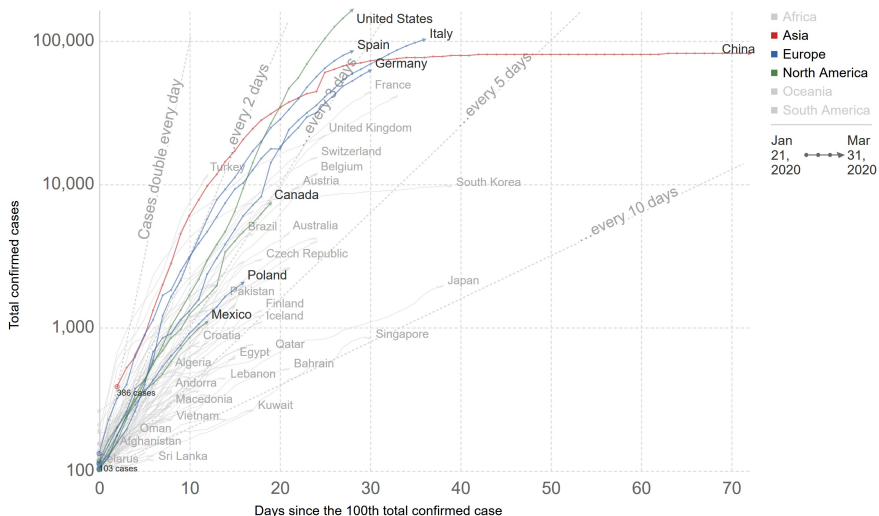
Total and daily confirmed COVID-19 cases per million people, United States

The number of confirmed cases is lower than the number of total cases. The main reason for this is limited testing.



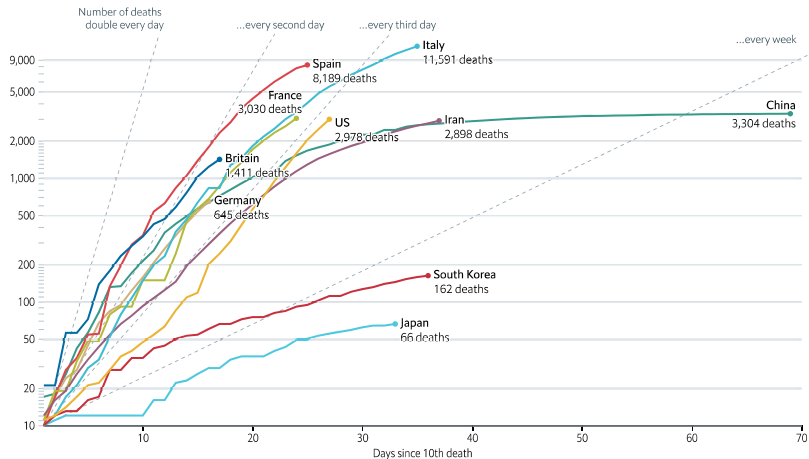
Total confirmed cases of COVID-19

The number of confirmed cases is lower than the number of total cases. The main reason for this is limited testing.



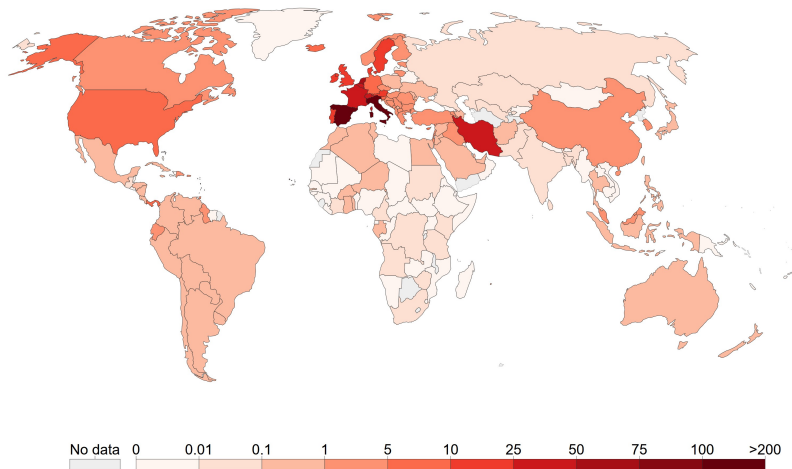
A deadly trajectory

Confirmed covid-19 deaths, to 10:56am GMT+1 March 31st, log scale



Total confirmed deaths due to COVID-19 per million people, Mar 31, 2020

Limited testing and challenges in the attribution of the cause of death means that the number of confirmed deaths may not be an accurate count of the true number of deaths from COVID-19.



Supply or Demand Shock? Very hard to identify!

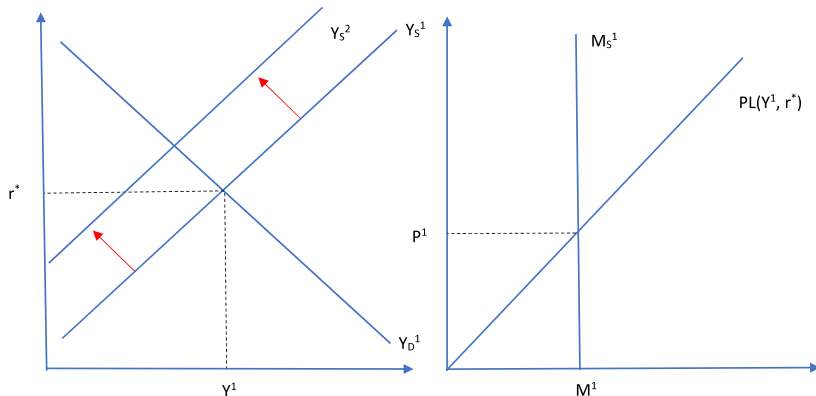
Why demand shock?

- ▶ People stop buying stuff, stop going out, stop travelling.
- ▶ Mainly affects services: restaurants, hotels, barbershops etc.
- ▶ Uncertainty about future dampens firm investment.

Why supply shock?

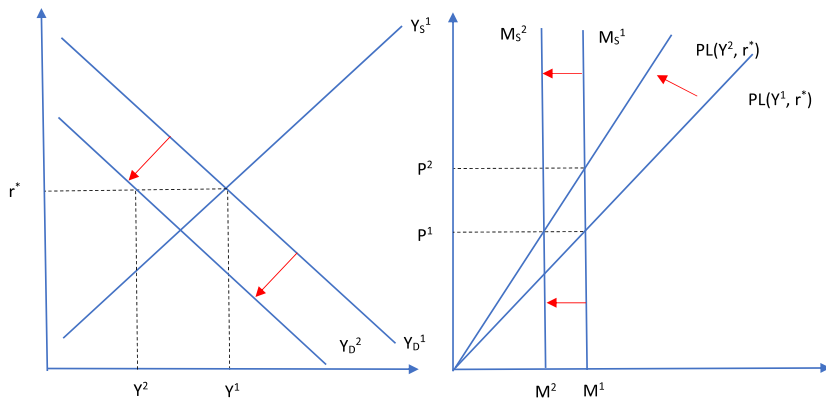
- ▶ People stopped working in many professions, negative labor supply shock.
- ▶ Distorted supply chains. Lack of parts for industries, broken global supply chains especially between Asia and Europe.

COVID-19 through the lens of the sticky price model



- ▶ Negative supply shocks shifts $Y_S^1 \rightarrow Y_S^2$.
- ▶ Output is demand driven here, still Y_1 . Economy produces above it's capacity (negative output gap).

COVID-19 through the lens of the sticky price model



- ▶ Negative demand shocks shifts $Y_D^1 \rightarrow Y_D^2$.
- ▶ With unchanged target interest rate r^* money supply has to go down, given prices are fixed at P_1 in the short run.

So what do we see in the data so far?

- ▶ Very limited hard data so far on the world economies response to the Coronavirus Pandemic.
- ▶ So far only short term indicators.
- ▶ Financial markets usually respond the fastest:
 - ▶ Stock market felt off the cliff
 - ▶ Uncertainty about the prospects of the economy rises. Recall Bush to Bernanke in 2008: "Are we heading to another Great Depression?"
- ▶ Slowly some hard data come up and they look terrifying. Recall Bernanke to Bush in 2008: "Looks like we are...", things look similar today...

Stock market tanks and wipes out all the gains since 2016



S&P 500 drops 20% in Q1

The S&P 500 recorded its worst quarter in more than a decade with its biggest drop since 2008.



Source: Refinitiv
Graphic: Tal Yellin, CNN

Uncertainty sharply rises

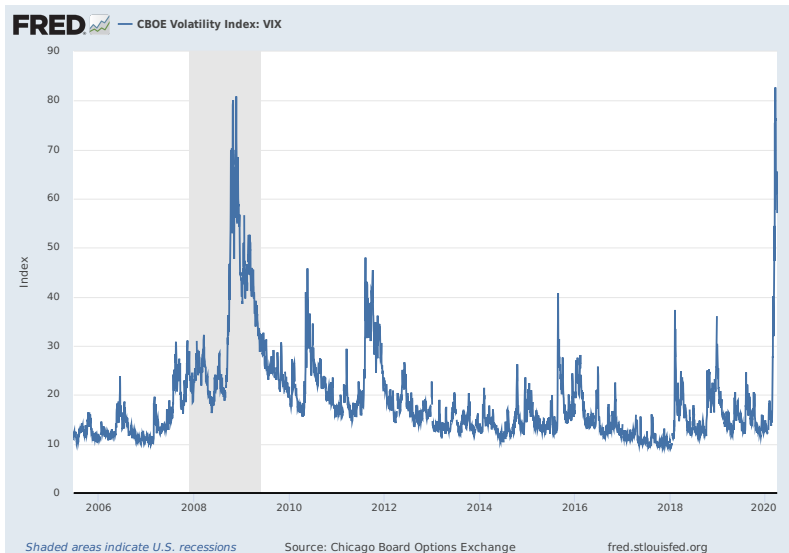
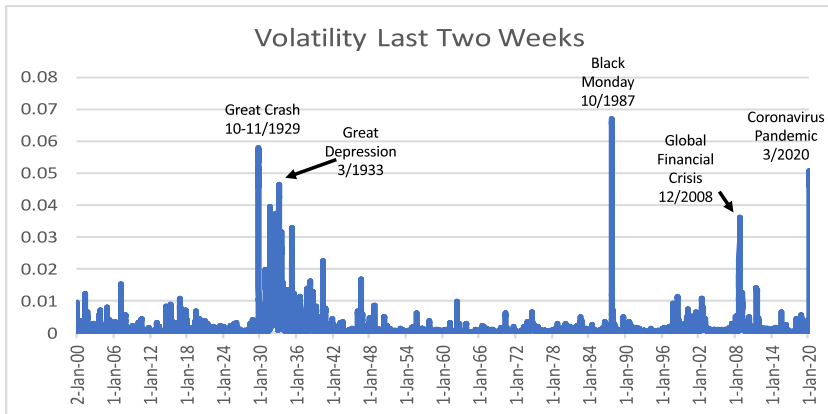


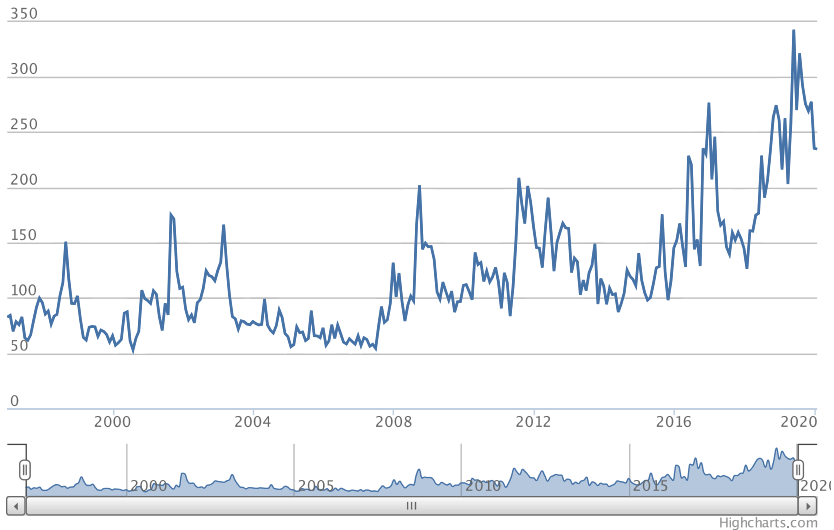
Figure 1. Realized U.S. Stock Market Volatility,
January 1900 to March 2020



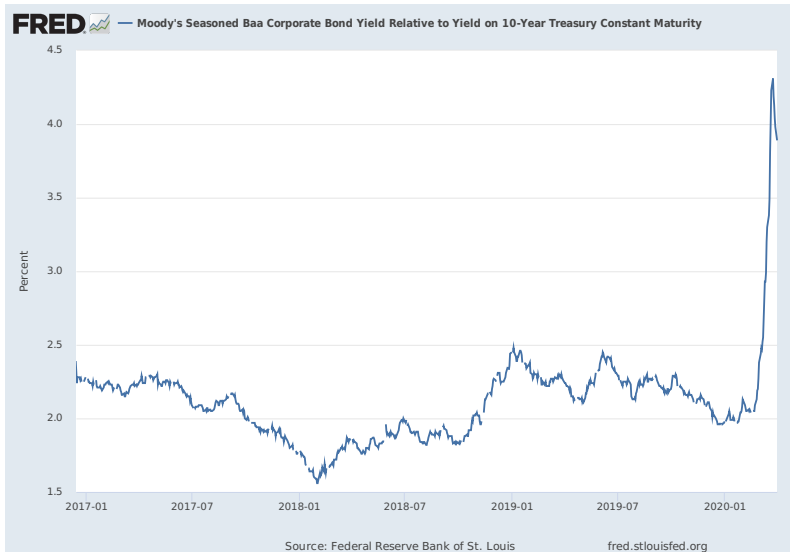
Notes: Sample period, 1/1/1900-3/23/2020. From 12/1925-Present, returns are computed using Yahoo Finance's 'adjusted close' series for the S&P 500 (^GSPC). Before that, returns are from the Global Financial Data extension of the Dow Jones Index. Volatility last two weeks is the sum of squared returns over the past 10 trading days.

Monthly Global Economic Policy Uncertainty Index

Zoom 1m 3m 6m 1y 7y All



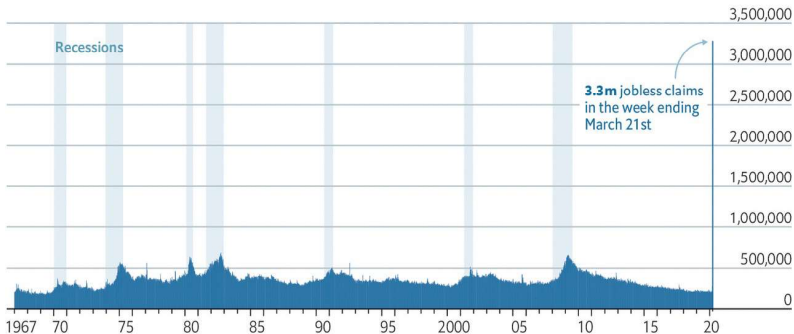
The spreads follow rising uncertainty



The labor market prospects look dire...

The coronavirus spike

United States, weekly initial unemployment claims



Sources: Department of Labour; NBER

The Economist

Macroeconomic Policy Response

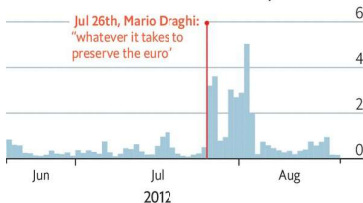
1. Classical monetary policy response: interest rate cuts
2. Fiscal stimulus plans - helicopter money.
3. Financial market interventions.
4. Liquidity provisions and debt reliefs for households and firms.
5. Labor market policy tools.

Policymakers learnt from the past experiences

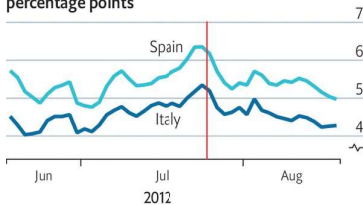
Talk is cheap

2012

Press mentions of "whatever it takes", '000

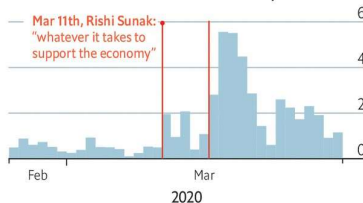


Ten-year government-bond spreads*, percentage points

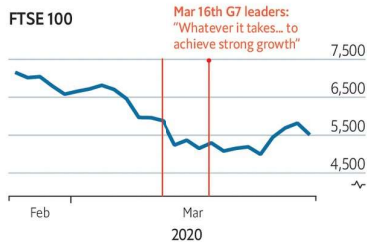


2020

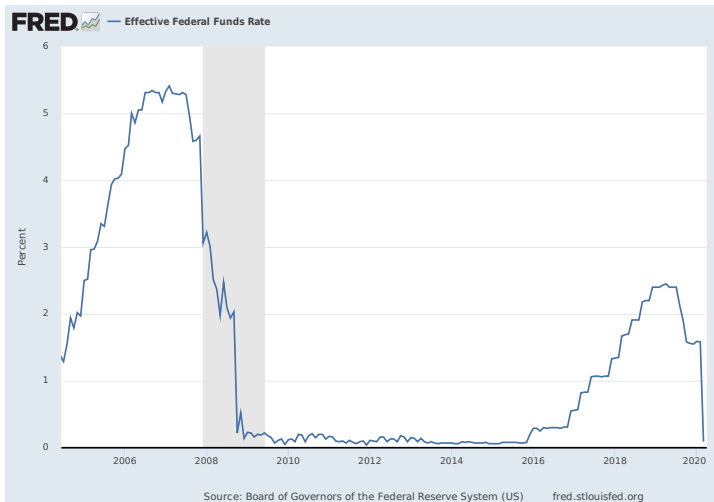
Press mentions of "whatever it takes", '000



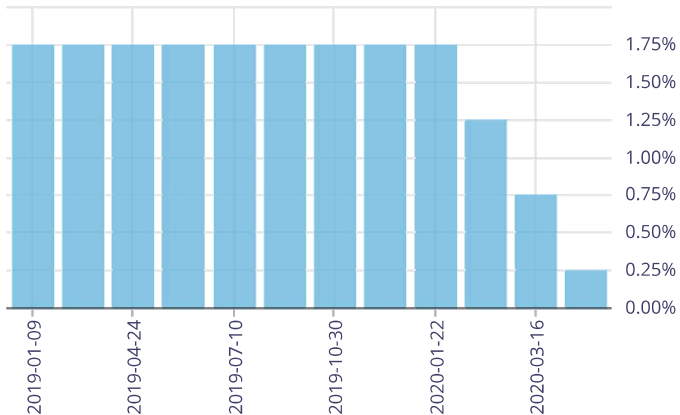
FTSE 100



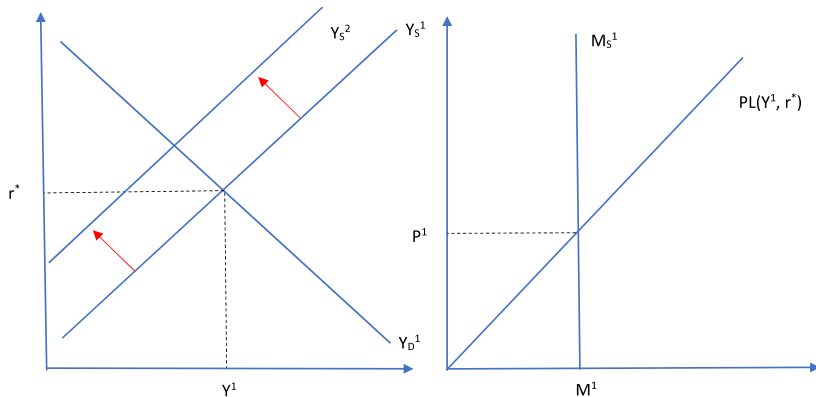
FED cuts the target interest rate ...



... so does the Bank of Canada

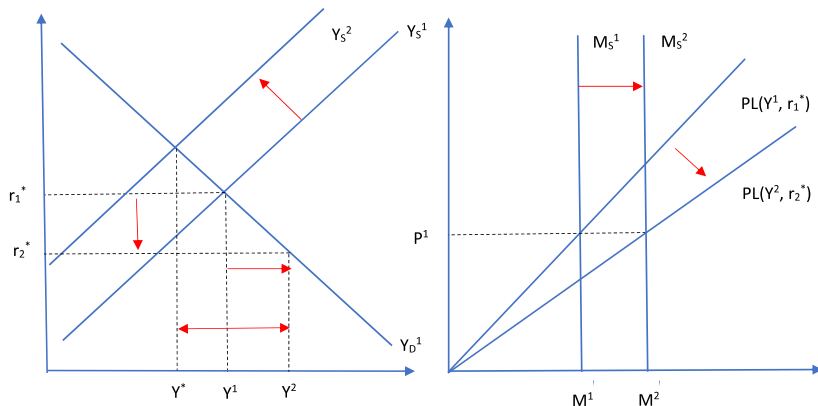


Monetary policy response to COVID-19 - supply shock



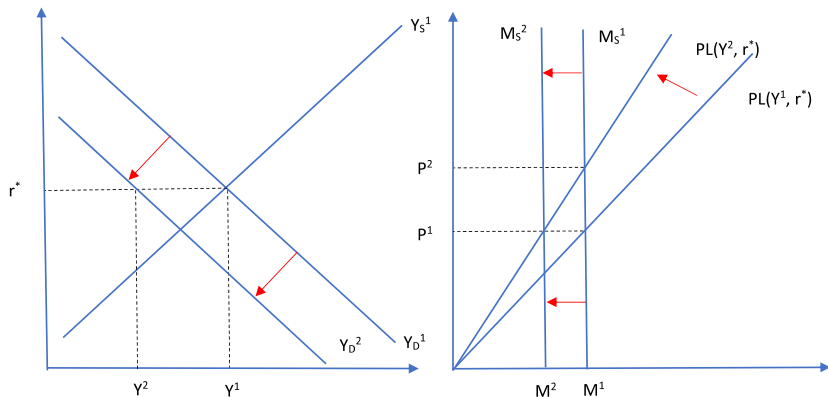
- Suppose first the shock is the negative supply shock.

Monetary policy response to COVID-19 - supply shock



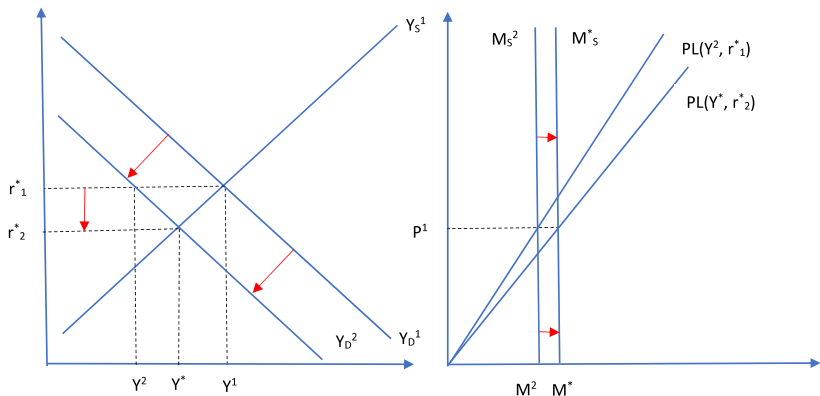
- ▶ Reducing the target interest rate $r_2^* < r_1^*$.
- ▶ Negative output gap rises! $(Y^* - Y_2) < (Y^* - Y_1) < 0$. The economy is overheated with shortages of supply!
- ▶ In the long run as prices adjust, higher inflation and low output.

Monetary policy response to COVID-19 - demand shock



- Suppose first the shock is the negative demand shock.

Monetary policy response to COVID-19 - demand shock



- ▶ Lowering the target rate to r_2^* restores efficiency but with lower output than the initial one, i.e. $Y^* < Y_1$.

US Fiscal Stimulus - \$2trn bazooka, one-tenth of GDP

1. \$250bn: sending cheques to Americans:

- ▶ Below some generous income thresholds (\$75,000 a year for a single person and \$150,000 for a married couple) every family can expect \$1,200 per adult and \$500 per child.

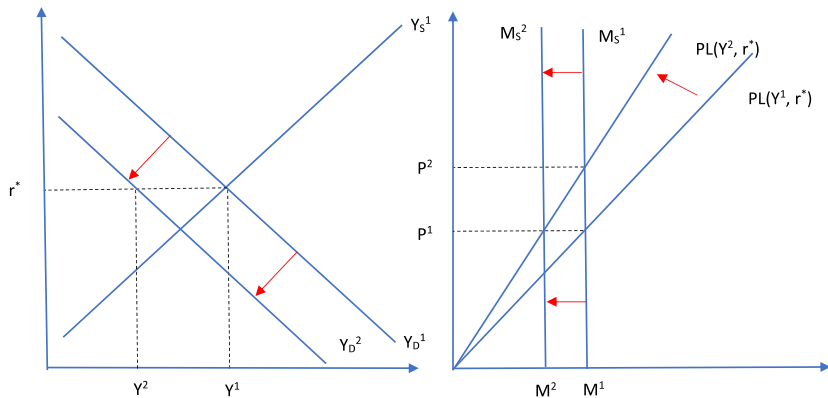
2. \$260bn: extending the safety net:

- ▶ The federal government would pay to top up unemployment-benefit levels by \$600 a week (current weekly average is \$385).
- ▶ Extended coverage: independent contractors, such as gig-economy workers. Those who have been laid off but not fired could receive compensation for lost hours.
- ▶ Unemployment insurance benefit period would be extended from the usual 26 weeks to 39 weeks.

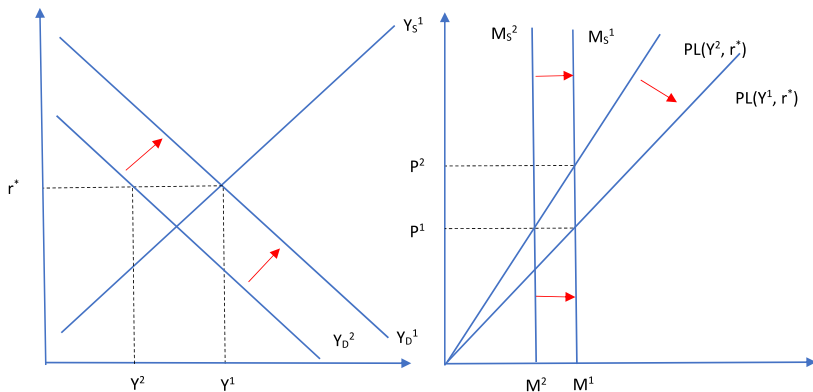
US Fiscal Stimulus - \$2trn bazooka, one-tenth of GDP

3. \$500bn: to stabilise firms and states through lending.
4. \$350bn: set aside to save small and medium-size firms (those with fewer than 500 employees)
 - ▶ loans of up to \$10m without interest or fees to pay for employees, rental costs and sundry other expenses.
5. \$75bn: set aside to bail out embattled big firms like airlines and those deemed critical to national security.

Fiscal Stimulus following the demand shock

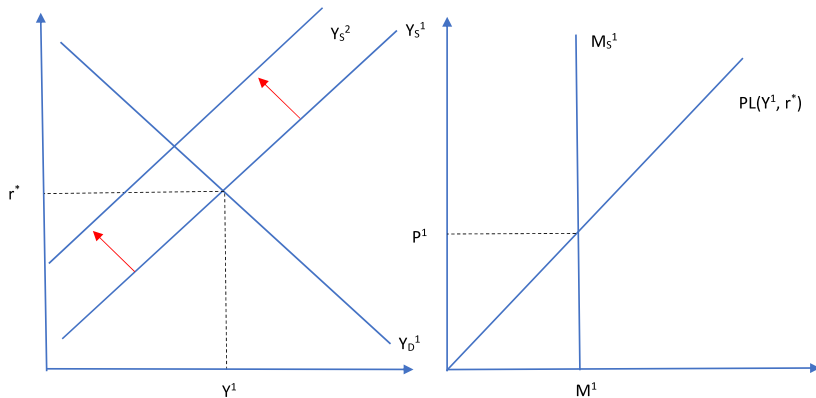


Fiscal Stimulus following the demand shock

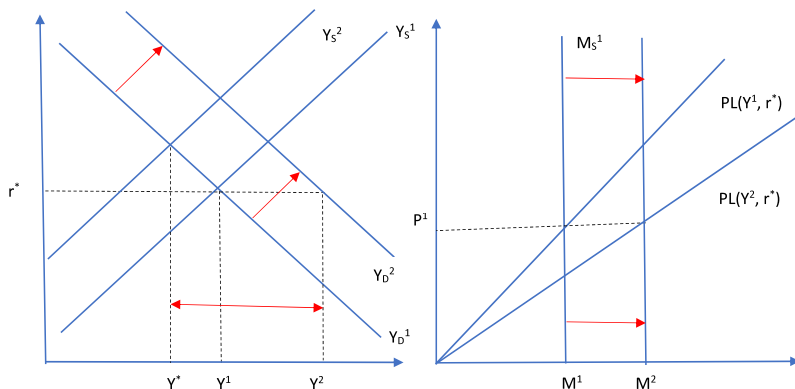


- ▶ Assume fixed labor supply upwards, i.e. intertemporal substitution of leisure is shut down (it's impossible to make people work harder with lockdown restrictions)
- ▶ Fiscal expansion can restore efficiency at the cost of crowding out private consumption - recall the logic of demand multiplier from Chapter 11.

Fiscal Stimulus following the supply shock

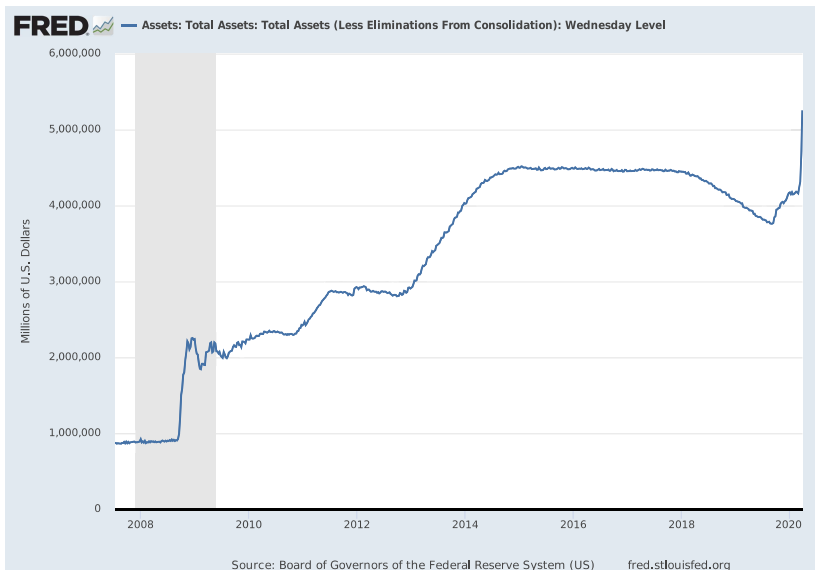


Fiscal Stimulus following the supply shock



- ▶ Fiscal expansion induces more negative output gap, the new output is now Y_2 . Economy is overheated, given shortages of the supply!
- ▶ Monetary policy has to accommodate here to keep the target rate unchanged. Both fiscal and monetary expansion in this case would make the problem worse and move the economy further away from efficiency.

FED acts as a lender of last resort



FED also designs it's own stimulus

"Aggressive effort must be taken across the public and private sectors to limit the losses to jobs and income and to promote a swift recovery once the disruptions abate,"

The stimulus measures include the following:

1. Unlimited purchasing of Treasury securities and mortgage-backed securities.
2. Lending up to \$300 billion to employers, consumers and businesses.
3. Establishing additional credit facilities to lend to large employers.
4. Increasing the flow of credit to municipalities.
5. Establishing a facility to lend to certain small and medium-sized businesses.
6. Purchasing investment-grade corporate bonds

Conslusions

- ▶ We are witnessing the beginning of the worldwide recession.
- ▶ Too early to say how deep and long it is going to be. Need more data and time.
- ▶ Very aggressive economic policies have been implemented early on. The lesson from the Great Recession has been learnt. Only time will tell if they are appropriate.
- ▶ Surely more supply oriented policies should help.
- ▶ Thanks to your hard work during the course you are equipped now to critically review the policies!